

Gemini Near Infra-Red Imager and Wavefront Sensor



Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Calibration and Maintenance Tools

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This document describes the software calibration programs and procedure for NIRI and the Near-Infrared OIWFS.

1 Introduction

1.1 Purpose

This document describes the software calibration programs and procedure for NIRI.

The intended audience of this document is:

- The NIRI reviewers;
- The developers of the Gemini Observatory Control System (OCS);
- Groups which will be using the design of the NIRI OIWFS (e.g. GNIRS);
- The maintainers of NIRI after it is delivered to Gemini.

1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.4 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 When is it necessary to recalibrate a mechanism?

2.1 What is the difference between normal calibration and low-level calibration.

Normal calibration is a procedure which must be carried out for many different reasons (listed below). It is often necessary: if the zero point of the analog input board drifts, if wires are replaced, if the mechanism is partially disassembled.

2.2 When is normal calibration necessary?

Normal calibration is necessary if a mechanism appears to be working and datums itself at the correct location, but the final verification phase (diagnose) fails. It is also necessary if the mechanism is moving to the correct positions, but returns an error when it moves.

The procedure for normal calibration is described in Section 5.3 and Section 6.4.

2.3 When is low-level calibration necessary?

Low-level calibration is necessary only if a mechanism has been disassembled and reassembled; appears to be operating correctly for the most part; and cannot datum itself or datums itself at an insane position. Low-level calibration should be considered only after all alternatives have been excluded. Note that only the focus stages and the gimbal mirror require recalibration. For all other mechanisms, all constants are well-defined by mechanical means.

2.4 When should the mechanism not be calibrated?

If the mechanism is not functioning properly, recalibrating it will mask problems or cause it to return false error messages.

3 Running the tools

3.1 The Engineering User Interface

You must know how to operate the NIRI / OIWFS Engineering user interface before you begin the calibration. See the software release notes [7] and the User's Guide for the Engineering User Interface [8] for more details.

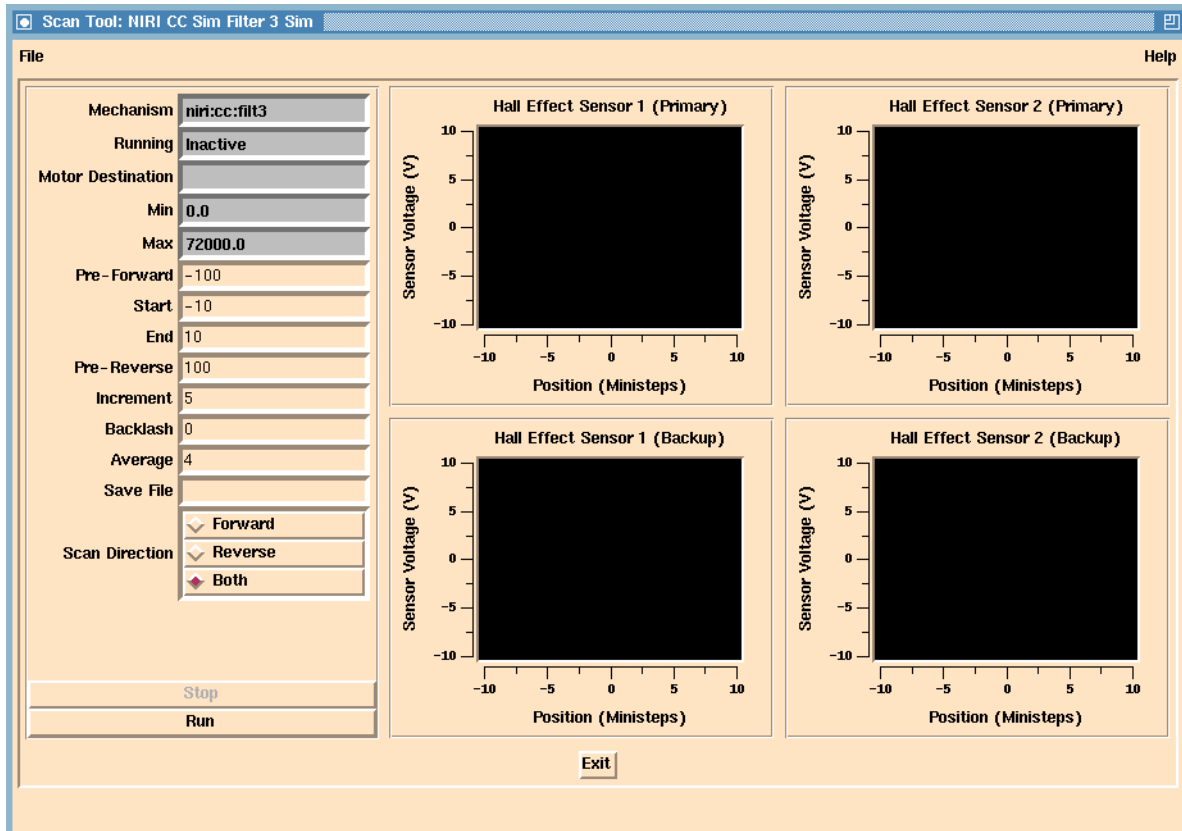
4 Scan Tool

The "scan tool" is used for characterizing hardware, for measuring backlash, and for diagnosing problems. It is a useful tool for diagnosing problems with mechanisms, particularly with mechanisms that are not moving freely. It can be helpful in low-level calibration of mechanisms. It is not used in normal operation.

4.1 Starting the “Scan Tool”

From the low-level engineering screen for the mechanism, press the “External Tools” button using the right mouse button, then select “Scan Tool.”

4.2 Using The “Scan Tool”



The normal mode of operation is as follows:

Forward Pass:

Move to “Pre-forward”.
 Move to “Start”, and read the Hall effect sensors
 Move to “Start” + “Increment”, and read the Hall effect sensors
 (Repeat until passes “End”)

Reverse Pass:

Move to “Pre-reverse”
 Move to “End”, and read the Hall effect sensors
 Move to “End” - “Increment”, and read the Hall effect sensors
 (Repeat until passes “Start”)

The difference between the values on the forward and reverse pass provides an indication of the amount of backlash in the system. Significant differences between the forward and reverse passes indicate a serious hardware problem, i.e., something is slipping or sticking.

5 The “Cycle Tool”

NIRI and the OIWFS are designed to provide quick verification that a mechanism is operating properly. This is done by placing analog Hall-effect sensors and magnets at various locations. When the mechanism is in a known, good state, the mechanism is moved to three test positions. The Hall-effect sensor values are read at each location.

The diagnose command verifies that the mechanism is operating correctly by moving it in sequence to each of these test positions and verifying that the sensor values are close to the which were recorded when the system was known to be operating correctly. The Cycle command is identical to the diagnose command, except that it repeats the test, until told to stop.

The “Cycle Tool” is a program that aids in this part of the calibration procedure. It moves the system to each of the test positions, and then records the values, which may then be written to a file. The file is read whenever the system is booted.

Warning: A side effect of running the “Cycle Tool” is that the in-memory table of expected values is changed. This is generally desirable, but can be misleading.

Warning: If the “Cycle Tool” is run while the mechanism is not working correctly then the results returned by the Cycle Test and Diagnose Test are suspect.

Note: The “Diagnose Test” is automatically performed when the “Datum” command is called from the high-level, interface. This is equivalent to the Datum/Diag command in the low-level engineering interface.

5.1 Starting the “Cycle Tool”

From the low-level engineering screen for the mechanism, press the “External Tools” button using the right mouse button, then select “Cycle Tool.”

5.2 Using the “Cycle Tool”

The screenshot shows a window titled "et_wish" with a menu bar (File, Mechanism, Tools, Help). The main area contains two columns of input fields for sensor data: V1 PA, V1 BA, V1 PB, V1 BB, V1 PC, V1 BC on the left and V2 PA, V2 BA, V2 PB, V2 BB, V2 PC, V2 BC on the right. Below these are fields for "Mech", ".pv base directory" (containing "/home/polaris/yamada/prj/xiri/ENG/pv"), ".pv subdirectory", "Ident", and "Out File". At the bottom are three buttons: "ReScan", "Write", and "Exit".

The fields / buttons of the cycle tool are:

V1PA. Voltage at test position A, sensor pair 1, primary sensor.

V1BA. Voltage at test position A, sensor pair 1, backup sensor

V2PA. Voltage at test position A, sensor pair 2, primary sensor.

V1BA. Voltage at test position A, sensor pair 2, backup sensor

V1PB. Voltage at test position B, sensor pair 1, primary sensor.

V1BB. Voltage at test position B, sensor pair 1, backup sensor

V2PB. Voltage at test position B, sensor pair 2, primary sensor.

V1BB. Voltage at test position B, sensor pair 2, backup sensor

V1PC. Voltage at test position C, sensor pair 1, primary sensor.

V1BC. Voltage at test position C, sensor pair 1, backup sensor

V2PC. Voltage at test position C, sensor pair 2, primary sensor.

V1BC. Voltage at test position C, sensor pair 2, backup sensor

Mech. The EPICS prefix for all fields related to this mechanism

.lut base directory. Where output files are created.

.lut subdirectory. Subdirectory of .lut base directory where files are created

Ident. The name of the default output file.

Out File. Name of the output file (user editable).

ReScan. Collect data.

Write. Write data file.

Exit. Exit from the “Cycle Tool.”

5.3 Normal Calibration (Phase 1)

1. Press the “ReScan” button to move the mechanism to the three test positions.
2. Examine the values. All values should be reasonable. (See Section 5.4.)
3. Examine the “Out File” field. Verify that the correct file will be created. Change the “Out File” field if you do not want to overwrite the normal file.
4. Press the “Write” button to write the data.
5. Press the “Exit” button to exit the “Cycle Tool.”
6. From a shell on the software server, change into the “.pv base directory” (from the “Cycle Tool”).
7. Run “gmake” in the “.pv base directory.”

Note: If you forget to press “Write” before pressing “Exit,” you will not be warned, but the file on disk will not be updated.

5.4 Expected Values (Phase 1)

The following tables describe the expected sensor values for each mechanism at the test positions.

The window cover has no sensors, and never needs to be calibrated. All sensor values read as 1 or 2 at the limits and 0 between limits.

TABLE 1.

Filter Wheels, The Pupil Mask Wheel, The Focal Plane Mask Wheel, The Entrance Beam Splitter, the Steering mirrors

Abbreviation	Sensor	Position	Significance	Value (V)
V1PA	1 (Primary)	A	First Position	< -0.5
V1BA	1 (Backup)			< -0.5
V2PA	2 (Primary)			0
V2BA	2 (Backup)			0
V1PB	1 (Primary)	B	Last Position	< -0.5
V1BB	1 (Backup)			< -0.5
V2PB	2 (Primary)			0
V2BB	2 (Backup)			0

The “Cycle Tool”

Abbreviation	Sensor	Position	Significance	Value (V)
V1PC	1 (Primary)	C	Home	> 1
V1BC	1 (Backup)			> 1
V2PC	2 (Primary)			0
V2BC	2 (Backup)			0

TABLE 2.

The Pupil Viewer

Abbreviation	Sensor	Position	Significance	Value (V)
V1PA	1 (Primary)	A	Alternate	< -1
V1BA	1 (Backup)			< -1
V2PA	2 (Primary)			0
V2BA	2 (Backup)			0
V1PB	1 (Primary)	B	Home	> 1
V1BB	1 (Backup)			> 1
V2PB	2 (Primary)			0
V2BB	2 (Backup)			0
V1PC	1 (Primary)	C	Home ^a	> 1
V1BC	1 (Backup)			> 1
V2PC	2 (Primary)			0
V2BC	2 (Backup)			0

a. If the corresponding values for positions B and C are not nearly identical, then the mechanism may have moved unexpectedly.

TABLE 3.

The Focus Stage

Abbreviation	Sensor	Position	Significance	Value (V)
V1PA	1 (Primary)	A	Large Positive	Near 0
V1BA	1 (Backup)			Near 0
V2PA	2 (Primary)			Any ^a
V2BA	2 (Backup)			Any
V1PB	1 (Primary)	B	Large Negative	Near 0
V1BB	1 (Backup)			Near 0
V2PB	2 (Primary)			Any
V2BB	2 (Backup)			Any
V1PC	1 (Primary)	C	Home	> 1
V1BC	1 (Backup)			> 1
V2PC	2 (Primary)			Any
V2BC	2 (Backup)			Any

a. The #2 sensor values should change significantly between the A, B, and C positions. Either $A > C > B$ or $B > C > A$.

TABLE 4.

The Gimbal Mirror

Abbreviation	Sensor	Position	Significance	Value (V)
V1PA	1 (Primary)	A	Large positive	Any
V1BA	1 (Backup)			Any
V2PA	2 (Primary)			Any ^a
V2BA	2 (Backup)			Any
V1PB	1 (Primary)	B	Large Negative	Any
V1BB	1 (Backup)			Any
V2PB	2 (Primary)			Any
V2BB	2 (Backup)			Any
V1PC	1 (Primary)	C	Home	Any
V1BC	1 (Backup)			Any
V2PC	2 (Primary)			Any
V2BC	2 (Backup)			Any

a. The #2 sensors should change significantly between the A, B, and C positions.

5.5 Configuration file

The test positions are defined in ENG/pv/cc.lut and ENG/pv/wfs.lut.

6 The “Position Tool”

6.1 Starting the “Position Tool”

From the low-level engineering screen for the mechanism, press the “External Tools” button using the right mouse button, then select “Position Tool.”

6.2 Using The “Position Tool”

NIRI and the OIWFS are designed to provide verification that a mechanism is in the proper configuration. This is done by placing analog Hall-effect sensors and magnets at various locations. When the mechanism is in a known, good state, the mechanism is moved to all standard positions. (For mechanisms with a continuous range of travel, the standard positions are at positions that are convenient for checking mechanism operation.)The Hall-effect sensor values are read at each position.

Each time that the mechanism is moved to a standard position, the expected sensor values are compared to the actual sensor values. If the values do not agree, then an error is returned.

Furthermore, any time that the mechanism is moved, whether to a standard position or not, the software examines the step count and sensor values, and attempts to determine if the mechanism is in a known position.

The “Position Tool”

The “Position Tool” is a program that aids in this part of the calibration procedure. It moves the system to each of the standard positions, and then records the values, which may then be written to a file. The file is read whenever the system is booted.

Warning: The “Cycle Tool” does not change the in-memory version of the table that it produces. To update the in-memory version of the table

1. Bring up the “Parameters” window (right-click the “Related Screens” button);
2. Install the updated table with gmake;
3. Press the “Load LUT” and “Load ILUT” buttons.

- 6.3 Warning: If the “Cycle Tool” is run while the mechanism is not working correctly, and if the values are written to disk, then the automatic verification of a move command will be suspect.

The screenshot shows a window titled "et_wish #2" with a menu bar (File, Mechanism, Tools, Help). Below the menu is a table with five columns: "count", "count tol", "val tol", "sens (p)", and "sens (b)". The table has 15 empty rows. Below the table are five input fields labeled "Mech", ".lut base directory", ".lut Subdirectory", "Ident", and "Out File". The ".lut base directory" field contains the path "/home/polaris/yamada/prj/xiri/ENG/pv". At the bottom of the window are three buttons: "ReScan", "Write", and "Exit".

Each line of this display represents a single defined mechanism position. The fields in each line are as follows:

count. Nominal step count as this standard position.

count tol. How close must the step count be to the nominal step count for the mechanism to be considered to be at this standard position.

val tol. How close must the Hall-effect sensor voltages be to the expected voltages for the mechanism to be considered to be at this standard position.

sens (p). Voltage at this standard position on the primary sensor.

sens (b). Voltage at this standard position on the secondary sensor.

Furthermore, the following values are defined.

Mech. The EPICS prefix for all fields related to this mechanism

.lut base directory. Where output files are created.

.lut subdirectory. Subdirectory of .lut base directory where files are created

Ident. The name of the default output file.

Out File. Name of the output file (user editable).

ReScan. Collect data.

Write. Write data file.

Exit. Exit from the “Cycle Tool.”

6.4 Calibration Procedure (Phase 2)

1. Press the “ReScan” button to move the mechanism to all defined positions.
2. Examine the values. All values should be reasonable. (See Section 6.5.)
3. Examine the “Out File” field. Verify that the correct file will be created. Change the “Out File” field if you do not want to overwrite the normal file.
4. Press the “Write” button to write the data.
5. Press the “Exit” button to exit the “Cycle Tool.”
6. From a shell on the software server, change into the “.pv base directory” (from the “Cycle Tool”).
7. Run “gmake” in the “.pv base directory.”

Note: If you forget to press “Write” before pressing “Exit,” you will not be warned, but the file on disk will not be updated.

6.5 Expected Values (Phase 2)

The following tables describe the expected sensor values for each mechanism at the standard positions.

The window cover has no sensors, and never needs to be calibrated. All sensor values read as 1 or 2 at the limits and 0 between limits.

TABLE 5.

Expected values

Position	Significance	Primary Value (V)	Backup Value (V)
pos00	First Position	any ^a	any
...	...	...	...
posNN	Last Position	any	any
home	Datum Position	> 1	> 1
park	Parked Position	any ^b	any

- a. These values should change significantly from position to position. If the same pair of values (within a few hundred millivolts) appears repeatedly, then the mechanism is not moving. If too many value pairs are near zero, then the mechanism is not moving correctly. If more than one positive value pair appears (other than the home or park position), then the mechanism is stuck at the home position.
- b. The park position is arbitrary, and is placed where it is most convenient. Its value might be positive, negative, or zero.

6.6 Configuration file

All constants that are defined in the position tool are defined in ENG/pv/cfg.tcl. See the comments in that file for more details.

7 Low Level Calibration Procedure

Whenever a mechanism is modified, it will be necessary to recalibrate it. Note that this procedure should only be used under very unusual conditions. It is not a standard maintenance procedure.

This section will describe the calibration procedure which is required for each mechanism.

7.1 The Wheels and Wheel-like Mechanisms, the Pupil Viewer, and the Window Cover

Low-level calibration is never necessary for the filter wheels, for the pupil-mask wheel, for the entrance beam splitter, for the focal-plane mask wheel, for the pupil viewer, for the window cover, and for the steering mirrors. There are no constants which require adjustment.

7.2 The Focus Stages

1. Start the Engineering Screen for the Science Focus Stage or OIWFS Focus Stage.
 2. Adjust the “New Position” field until the value of sensor 2P is less than 0.005 V. Call this *position A*.
 3. Repeat for sensor 2B. Call this *position B*.
 4. While observing the hardware, adjust the “New Position” field until the sensor is adjacent to the magnet. Call this *position C*.
 5. Compute the value of *position C* - *position A*. This is the value of *DOFP*.
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6. Compute the value of *position C - position B*. This is the value of *DOFB*.
7. Bring up the parameter window. In the *DOFP* field, enter the value of *DOFP*. In the *DOFB* field, enter the value of *DOFB*. Close the parameter window.
8. Press the Datum Button, and let the system datum.
9. Set the “New Position” field to 0. You should find that sensor 1P has a signal between 1 and 5 V. If this is not true, then there is a hardware problem or the calibration was performed incorrectly.
10. While observing the stage, adjust the “New Position” field until you know the safe limits of motion of the mechanism. It is better to be slightly conservative here. Call the lower limit *LOPR*. Call the upper limit *HOPR*.
11. Bring up the parameter window. In the *HOPR* field, enter the value of *HOPR*. In the *LOPR* field, enter the value of *LOPR*.
12. Edit *niri/ENG/pv/cfg.tcl*. Enter the value of *LOPR* in the *pos01* field for *ccFoc*. Enter the value of *HOPR* in the *pos02* field for *ccFoc*.
13. Run Cycle Tool.
14. Run Position Tool.
15. Bring up the parameter window. Enter 0 in the *DOF1* field.
16. Disable sensor 1P.
17. Press the Redatum button in the engineering window. You should eventually see “Peak at (some value)”. Multiply this value by -1 and enter it into the *DOF1* field.
18. Enter sensor 1P.
19. Calibration is complete. The values which were placed in the *DOF1*, *DOFP*, and *DOFB* fields should be entered into *ENG/pv/cc.pv* as the values for *focHallstep.DOF1*, *focHallstep.DOFP*, and *focHallstep.DOFB*.
20. Enter the lower limit in *ENG/pv/cc.pv* as the value of *focHallstep.LOPR* and *focHallstep.CDSB*. Enter the upper limit as *focHallstep.HOPR* and *focHallstep.CDSA*.
21. Run *gmake*, then reboot the IOC.

7.3 The Gimbal Mirror

This section has not been written yet.

8 Warnings

If the system needs to be recalibrated frequently, there is a problem with the mechanism. Recalibration can mask minor problems, particularly with mechanisms that are not moving freely.